Closure-Based Outer-Frame Strengthening for Simple Critical Zeros

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Abstract

We strengthen the outer case of a fixed nine-point certificate for simple critical zeros of the Riemann zeta function. A different baseline-only dual representation pays 201 captured frames; 35 previous frame witnesses are retained. Exact difference-constraint closure and seventeen closed cases pay the remaining five frames. The original 2,399 ordered pair domains are all replayed at the new uniform reward $805460/10^8$. The unchanged all-zero lossless transport gives the asymptotic lower proportion $66812491/99194540 = 67.3550086526939\ldots\,\%$ for both simple and distinct critical zeros, relative to all nontrivial zeros counted with multiplicity. The new mathematical certificate uses only original proved point and convexity data. It is separate from the completed Lean and independent-kernel verification of the preceding bound; no new formal or website acceptance is claimed here.

1 Statement and retained framework

Let $N(T)$ count all nontrivial zeros of ζ in $(0,T]$ with multiplicity, and let $N_s(T)$ count its simple zeros on $\Re s = 1/2$ in that interval. Let $N_*(T)$ count distinct critical zeros. Put

$$p = \frac{66812491}{99194540}, \qquad c = \frac{805460}{10^8}.$$

Theorem 1.1. *In the unconditional correlation and continuous-certificate framework retained in [1, 2, 3],*

$$\liminf_{T\to\infty} \frac{N_s(2T) - N_s(T)}{N(2T) - N(T)} \geq p, \qquad \liminf_{T\to\infty} \frac{N_s(T)}{N(T)} \geq p.$$

The same statements hold with N_s replaced by N_*. Equivalently, each ratio is eventually at least $p - \varepsilon$ for every $\varepsilon > 0$.

The denominators retain all line-exterior zeros and every multiplicity. These are asymptotic statements, not a fixed-height numerical count. Relative to the preceding bound $p_0 = 66812491/99194552$, the gain in percentage points is exactly

$$100(p - p_0) = \frac{1002187365}{122994474451826} = 0.00000814823080034\ldots.$$

The purpose is a complete net improvement within a fixed certificate, rather than a new correlation theorem.

We retain the fixed positive normalized AM density and its real Fourier transform K from [3]. Write $w(t) = K(t)^2$, so $0 \leq w \leq 1$. With the unchanged nonnegative pair coefficients a_{ij} and gap pressures b_r recorded there, set

$$F_8(h) = \sum_{r=0}^{6} b_r h_r + \sum_{0\leq i<j\leq 7} a_{ij} w(h_i + \cdots + h_{j-1}), \qquad h_r \geq 4/5.$$

For eight such gaps define

$$F_9(g) = \tfrac{1}{2}F_8(g_0,\dots,g_6) + \tfrac{1}{2}F_8(g_1,\dots,g_7) + 2w(g_0+\cdots+g_7).$$

The unchanged strict strong capture at $\tau = 805803/10^8$ covers every frame with $F_8 < \tau$ by 241 original cells and their reflections. Compatible ordered pairs are covered by the retained 2,399 difference-constraint domains. Thus no additional captured leaf or analytic table is assumed.

2 Complete payment of the outer case

Set $\ell = 805117/10^8$. The new certificate gives the following continuous implication, rather than just values at sampled gaps.

Lemma 2.1. *Every real frame with $h_r \ge 4/5$ satisfies $F_8(h) \ge \ell$. Every compatible pair of captured frames satisfies $F_9 \ge M$, where*

$$M = \frac{32992055894668884928258784247}{40960000000000000000000000000000} > c.$$

Proof. For the original 241 cells we use 201 distinct baseline-only dual witnesses, 35 retained frame replacements from [3], and five new closed covers described below. The first 201 witnesses use exactly the original 33-variable rows, with no new anchor. They are the frozen adapter data accompanying the completed formalization [4]; their real inequalities are now recomputed at ℓ. They are not the old YA multiplier vectors.

The five new labels are $17, 69, 74, 131, 183$. In the following table $S_{ij} = h_i + \cdots + h_{j-1}$ and $SC = 32768$. Every branch is closed; successive endpoints coincide, and the first and last endpoints are the exact whole-cell closure bounds. Hence their union covers the complete cell, including the branch boundaries.

Label	Span	$\min(5SCS_{ij})$	$\max(5SCS_{ij})$	Interior cuts
17	$(6,7)$	282370	341840	297237, 312105, 326972
69	$(6,7)$	275775	375395	300680, 325585, 350490
74	$(0,7)$	1745465	1836125	none
131	$(6,7)$	292790	382240	315152, 337515, 359877
183	$(1,2)$	273900	324685	286596, 299292, 311988

Each branch closes the actual eight-node difference constraints again. Its seven gap variables and all 26 square variables retain complete real boxes, including $0 \le w \le 1$. Added rows comprise exact closure bounds and valid tangents from the original point table and convexity regions. Seventeen nonempty cases use 387 additional rows, 529 positive dual multipliers, and 288 distinct original point entries. They replace whole cells, not selected neighborhoods of numerical minimizers.

All 253 payments, $201+35+17$, are evaluated by exact rational arithmetic. Their minimum is attained at label 17, case index 2:

$$L = \frac{82444006766814702334887673 97}{10240000000000000000000000000000}, \qquad L-\ell = \frac{2596681470233488767397}{10240000000000000000000000000000} > 0.$$

The gap pressures and pair coefficients are reflection invariant, so the same lower bound covers all 241 reflected cells. Outside the strict capture, $F_8 \ge \tau > \ell$. This proves the first claim.

For the second claim we rebuild all compatible domains and replay all their real boxes and exact residuals. The retained family has 2,342 original pair witnesses, 47 earlier replacements with 84 closed cases, and ten replacements from [3] with twenty closed cases. None is replaced by a stored summary of its minimum. Their exact minimum remains M, attained at $(406, 70)$, and

$$M - c = \frac{41429466888492825878424 7}{40960000000000000000000000000000} > 0.$$

□

Lemma 2.2. *Every real separated nine-point frame satisfies $F_9(g) \geq c$.*

Proof. If both eight-point subframes lie in the strict low set, the complete pair cover and Lemma 2.1 give $F_9 \geq M > c$. Otherwise one subframe has $F_8 \geq \tau$ and the other has $F_8 \geq \ell$. Nonnegativity of the total-span square gives

$$F_9 \geq \frac{\tau + \ell}{2} = \frac{805460}{10^8} = c.$$

The last safe bound equals c; it is not asserted to be strictly larger. The stored full-frame minimum L additionally gives a strict positive outer margin of $(L - \ell)/2$. □

3 Validity and all-zero transport

We recall why the finite witnesses imply real inequalities. If $v \leq w(q)$, $d_- \leq w'(q) \leq d_+$, and w is convex on $[l, u]$ containing both the anchor q and the entire physical span interval, then

$$w(x) \geq v + d_+(l - q) + d_-(x - l),$$
$$w(x) \geq v + d_-(u - q) + d_+(x - u).$$

This includes the derivative-interval correction and remains valid when closure puts the anchor outside the physical span interval. Every newly used point pack and enclosing region is from the original proved PTL/REG source. New derivative sampling or a strong-convexity assumption is not used.

For a branch write its actual inequalities as $Ax \leq b$ and $L_i \leq x_i \leq U_i$, with objective numerator $q^T x$. A rational multiplier vector $\lambda \geq 0$ gives, with $r = q + A^T\lambda$,

$$q^T x \geq -\lambda^T b + \sum_i r_i \begin{cases} L_i, & r_i \geq 0, \\ U_i, & r_i < 0. \end{cases}$$

Every signed residual is paid. Floating optimization only proposes the witnesses; the implication, closure, region guards, complete boxes and closed unions are checked with integers and fractions.

The lossless counting transport of [1, 2] applies to Lemma 2.2 with unchanged gain $A = 67216841/10^8$ and gap fee $B = 404350/10^8$. Sliding retains pair budget at most two, pair mass below sixteen, endpoint fee $8c$, a close-pair reward at least $(51/550)^2 > c$, and a continuous majorant of mass below two. Its smoothing loss remains $32d\varepsilon$. The Hermitian Gram argument includes reflected off-line pairs and all zero multiplicities. Taking the height limit first and the smoothing limit second gives

$$\frac{A - B}{1 - c} = \frac{66812491}{99194540} = p.$$

The same retained transport proves the dyadic and cumulative versions. Simple-zero inclusion gives the distinct-zero version, completing Theorem 1.1.

4 Reproduction and proof status

The additive public certificate and standard-library checker are

`output/am-outer-frame-certificate.json`
`scripts/am_outer_frame_certificate.py`.

Run the checker from the repository root with `--check`. It rebuilds all 253 original frame cases, verifies reflection symmetry, regenerates compatible geometry, and replays all 2,399 ordered pair witnesses. The 201 baseline adapter is included in the public JSON, so an external formal-repository checkout is not needed. Optional `--source` verifies actual original point and region identity. Optimized Python is explicitly rejected.

The preceding bound p_0 has a complete ordinary Lean proof and an independent full-kernel replay of 84,362 declarations at the frozen commit [4]. This paper proves a further mathematical improvement, with a separate exact finite check; its new witnesses have not yet been transplanted into Lean. The older formal records, submissions, papers, and certificates are retained. Website resource compliance and acceptance remain separate pending tasks.

References

[1] Li Xiang, *Lossless Eight-Point Assembly for Simple Critical Zeros of the Riemann Zeta Function*, 2026. Retained source and proof scope in RH-Weil.

[2] Li Xiang, *Complete Payment of the Ninth-Span Square for Simple Critical Zeros*, 2026. Retained in RH-Weil.

[3] Li Xiang, *Joint Frame and Pair Strengthening for Simple Critical Zeros*, 2026. The preceding complete certificate and its original continuous-capture dependencies.

[4] Li Xiang, RH-Zero-Proportion-Formalization, commit 14362fe, complete JointFramePair Lean proof and independent nanoda replay, 2026.